

Anchoring Large Language Models to Empirical Behavioral Evidence

A Hybrid Approach for DCE-Grounded Narrative Policy Simulation

Blinded for peer review

Abstract

Objective: To develop and validate empirical-frontier regularization (EFR), an inference-time procedure anchoring LLM outputs to a DCE-estimated reference surface without retraining.

Methods: A Wuhan vaccination DCE ($N=1,027$) was combined with LLM simulation on a 64-state policy grid. EFR computes $\pi_{\text{EFR}} = \lambda \bar{\pi}_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$ with $\lambda = 0.25$ fixed as a conservative operating point.

Results: Held-out validation on six DCE choice tasks ($N = 205$ respondents) shows multinomial log loss: Pure-DCE 0.911, EFR 0.923, Raw LLM 1.020. EFR reduced log loss by 9.5% versus Raw LLM and approached Pure-DCE performance (within 1.3%). The unconstrained LLM violated waiting-time monotonicity on 62.5% (Qwen2.5-72B), 34.4% (DeepSeek V4 Pro), and 10.4% (MiroThinker) of scenarios; after EFR, MVR-Wait dropped to 30.2%, 7.3%, and 8.3% respectively. Rank correlation ρ rose from 0.061/0.302/0.531 (Raw) to 0.689/0.812/0.756 (EFR), all exceeding the uniform-noise null ($\rho = 0.367$, 95% CI [0.145, 0.563]); two of three models also exceeded the stricter permutation null that preserves the LLM’s marginal distribution (Supplement B).

Limitations: Single-site DCE; respondent-level holdout evaluates generalization to unseen respondents, not unseen tasks. Out-of-design scenarios lacked human ground truth.

Conclusion: EFR constrains LLM outputs to a DCE-derived frontier, bounding deviation while preserving narrative-extension capability. It is a wrapper for existing DCEs, not a replacement.

Keywords: empirical-frontier regularization; large language models; discrete choice experiment; health technology assessment; behavioral simulation

Highlights

1. EFR is an inference-time regularization layer that constrains LLM outputs to a DCE-estimated reference without retraining.
2. On held-out respondent validation, EFR recovers most of the DCE model's predictive performance while retaining a bounded LLM component.
3. EFR improves structural consistency (monotonicity, rank order) but does not establish behavioral validity for out-of-design narrative scenarios.

Article Type: Methodological Article

1 Introduction

Large language models (LLMs) are increasingly used as simulated decision-makers for health-care policy^{19,18,27}. Their appeal is that they can interpret rich natural-language scenarios that conventional structured decision models cannot. However, linguistic fluency does not guarantee behavioral reliability: an unconstrained LLM may assign higher acceptance to a longer-waiting policy than to an otherwise identical shorter-waiting policy, contradicting empirically observed preference patterns and producing unreliable policy rankings.

Current approaches do not explicitly enforce DCE-estimated behavioral constraints^{1,12,20}: LLM agents may produce decisions that appear linguistically plausible while violating basic preference relationships implied by observed human behavior.

One source of such constraints is the discrete choice experiment (DCE), which estimates how observed human preferences vary with decision attributes⁵. In this study, we treat DCEs not as models to be replaced by LLMs, but as sources of DCE-estimated behavioral reference. A DCE-estimated choice frontier defines the preference ordering and directional relationships supported by observed responses. The challenge is to combine this empirical behavioral knowledge with the representational flexibility of LLMs, so that simulated predictions can incorporate an LLM-derived narrative component while remaining closer to DCE-implied structure; whether the resulting narrative extension is behaviorally more accurate on out-of-design scenarios is not established by the present study, because no human ground truth is available outside the original DCE design space.

We propose a **hybrid approach** that anchors LLM-generated choice probabilities to empirical behavioral evidence from DCEs. The method estimates an empirical choice frontier from DCE responses and blends the LLM’s raw probability with the frontier through a transparent weighted-averaging procedure. The adjustment is auditable: the raw LLM probability, the empirical frontier value, the blending weight, and the final adjusted probability are all directly inspectable. The method requires no retraining. The same LLM can be anchored using different empirical datasets, enabling domain-specific grounding while preserving general capabilities.

We evaluate the approach using a real-world vaccination-timing DCE involving 1,027 adults in

Wuhan, China. This study makes three contributions:

1. We introduce a lightweight inference-time adjustment procedure for grounding LLM-based simulation in empirically estimated human behavioral knowledge.
2. We show that the proposed adjustment improves structural consistency across multiple LLMs, reducing behavioral inconsistencies and improving agreement with the DCE-estimated reference. On held-out-respondent validation (local Ollama Qwen2.5-72B replication), EFR materially improved predictive performance relative to the unconstrained LLM (9.5% reduction in log loss) and approached the Pure-DCE model (within 1.3%), while retaining a non-zero LLM component. **The retention of an LLM component is motivated by narrative-extension capability, not by evidence that LLMs provide accurate behavioral simulation for novel scenarios; the OOD evaluation demonstrates constraint effects, not validation of LLM behavioral accuracy.**
3. We demonstrate that empirically grounded behavioral models can be extended to richer natural-language healthcare policy scenarios while preserving auditable behavioral constraints.

2 Related Work

Our work connects two research areas: LLM-based synthetic agents for healthcare simulation and empirical behavioral models derived from discrete choice experiments (DCEs). While these two directions have developed largely independently, both seek to improve the realism and reliability of decision support. Our work lies at their intersection by incorporating empirically estimated behavioral knowledge into LLM-based synthetic agents through inference-time regularization.

Prior LLM-for-HTA work in this journal. Recent issues of *Value in Health* have featured LLMs along trajectories distinct from the behavioral-simulation axis we pursue: evidence extraction²², sponsor-HTA conversation simulation¹⁷, microsimulation²⁴, and LLM-based prediction of stated preferences⁶. We deploy LLMs as narrative-extension mechanisms anchored to a DCE-derived

empirical frontier, not as tools for review or negotiation. The closest analogue outside HTA is Twin-2K-500²⁵, which validates LLM-as-consumer simulation in marketing; we extend this line to health policy. A pre-existing LLM-powered social digital twin framework¹⁴ is conceptually adjacent but does not anchor to a pre-collected empirical choice frontier.

2.1 LLM-Based Synthetic Agents

Large language models (LLMs) have enabled a new generation of synthetic agents capable of generating natural-language reasoning and simulating human decision making across diverse domains^{3,19,18}. In healthcare, these agents have been explored for public health communication, policy evaluation, and behavioral simulation^{21,7}. Their principal advantage is representational flexibility: unlike conventional statistical models, LLMs can process rich textual descriptions and respond to scenarios that extend beyond predefined structured variables.

However, behavioral realism remains a major challenge: LLM responses may violate expected directional relationships (e.g., waiting-time monotonicity, where longer waits should reduce acceptance) or produce implausible preference orderings^{1,12,11}. Prompt engineering can reduce some inconsistencies but provides no explicit mechanism for enforcing behaviorally grounded constraints^{29,28}; the mechanism cannot guarantee compliance across diverse narrative phrasings outside the original prompt template. We adopt the LLM-as-narrative-component framing of Argyle et al.², Horton¹²: the LLM’s narrative output is one input to a convex blend with an empirical choice frontier, not a stand-alone behavioral model.

2.2 Empirical Behavioral Models from Discrete Choice Experiments

Discrete choice experiments (DCEs) provide a well-established framework for estimating how individuals trade off policy-relevant attributes such as waiting time, vaccine efficacy, side effects, and financial incentives^{23,5}. Under the random utility framework^{26,16}, DCE responses can be used to estimate an empirical choice frontier that preserves interpretable relationships between decision attributes and observed choices. Mixed-logit models further account for preference heterogeneity

across respondents^{9,10}.

In this study, we treat the DCE frontier as an empirically estimated source of behavioral knowledge rather than as assumption-free ground truth. While empirical choice models provide transparent and behaviorally interpretable predictions within their original experimental design, they cannot directly operate on rich natural-language policy scenarios.

2.3 Bridging Empirical Behavioral Models and LLM-Based Agents

Existing LLM-based synthetic agents provide representational flexibility but lack mechanisms for enforcing empirically estimated behavioral constraints^{15,24}. DCE-derived behavioral models provide transparent and interpretable structure but remain confined to their experimental design. Existing approaches trade flexibility for reliability or vice versa.

3 Methods

This study integrated two methodological components: (1) a discrete choice experiment to estimate empirical vaccination preferences, and (2) a procedure for aligning LLM predictions with these empirical patterns. The following sections describe each component and the validation approach.

3.1 Overview

Section 3.2 describes the DCE design and empirical frontier; Section 3.3 presents the EFR framework. Section 4.2 reports the 64-state evaluation, and Section 4.3 reports the out-of-design narrative evaluation; Section 5.6 concludes.

3.2 Stage 1: Empirical Behavioral Knowledge Extraction

The first stage extracts empirical behavioral knowledge from a vaccination-timing DCE conducted among 1,027 adults in Wuhan, China (fielded December 2022–January 2023). This knowledge is

an empirical choice frontier that serves as the behavioral reference for inference-time regularization.

DCE design and sampling. The Wuhan vaccination-timing DCE used a **non-orthogonal fractional design** with five policy-relevant attributes: waiting time (0, 1, 3, 6 months), vaccine efficacy (50%, 70%, 95%), side-effect risk (0.1%, 1%, 10%), cash incentive (50, 200, 800 RMB), and vaccine origin (Domestic, Imported). The intended non-opt-out candidate space contained 216 profile cells ($4 \times 3 \times 3 \times 3 \times 2$). All 1,027 respondents completed the **same six choice tasks**, each presenting two non-opt-out alternatives and an opt-out alternative; the design therefore contained six unique administered choice tasks covering 12 of the 216 candidate cells (5.6% coverage). Task presentation order was randomized across respondents to mitigate order effects. The design was generated using D-efficiency criteria with prior parameter estimates from pilot data; a post-hoc dominance audit confirmed that none of the six administered A-versus-B profile pairs exhibited strict dominance across the monotonic attributes.

Sampling and ethics. Respondents were recruited through an online panel provider (quota sampling: age 18–65, stratified by gender and education to approximate Wuhan adult population distribution). Eligibility criteria: adult resident of Wuhan, no prior COVID-19 vaccination. The full DCE questionnaire and recruitment protocol are available in the public repository. Individual-level DCE responses are not publicly released because of participant confidentiality. The study was reviewed and approved by the relevant institutional ethics committee, and all participants provided informed consent before participation.

Stakeholder involvement and instrument design. Two physician interviews and two rounds of twelve cognitive interviews (ISPOR good-research-practices^{13,4}) refined the choice-task framing.

Construction of the Empirical Behavioral Reference. The empirical behavioral reference was constructed from a 6-parameter grouped/conditional logit with five attribute coefficients and an opt-

out alternative-specific constant, estimated on the Wuhan DCE. The frontier maps each candidate policy to a DCE-implied binary policy-versus-opt-out probability (i.e., the probability of choosing the policy alternative over the opt-out under the conditional logit model), providing a deterministic signal that anchors the LLM’s narrative choices.

3.3 Stage 2: Inference-Time Adjustment Procedure

The adjustment algorithm aligns LLM outputs with the empirical reference by computing $\pi_{\text{adj}}(y = 1 \mid \mathbf{x}_g) = \lambda \cdot \pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g) + (1 - \lambda) \cdot P_{\text{emp}}(y = 1 \mid \mathbf{x}_g)$ (the EFR equation $\pi_{\text{adj}} = \lambda \pi_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$; full derivation in Appendix A.1 of the Supplementary Material). This binary policy-versus-opt-out formulation is the primary definition; for the exact-task categorical validation, EFR is extended to a convex blend of full three-alternative probability vectors (Appendix N of the Supplementary Material).

The final stage applies the adjusted probability to policy evaluation, comparing policy scenarios and estimating changes in expected choice probability.

For a baseline scenario $\mathbf{x}$ and an intervention that changes attribute x_k to x'_k , the policy response is:

$$\Delta_k(\mathbf{x}) = \pi_{\text{adj}}(y = 1 \mid \mathbf{x}') - \pi_{\text{adj}}(y = 1 \mid \mathbf{x}),$$

where $\mathbf{x}'$ denotes the post-intervention scenario. Averaging over comparable scenarios gives the mean policy response for attribute k .

Because evaluation operates only on the adjusted probability surface, the approach can be deployed locally without transmitting individual-level DCE responses to external providers. The resulting policy-response quantities are descriptive summaries and should not be interpreted as causal treatment effects.

Table 1: Behavioral consistency with the DCE-estimated reference before and after EFR on the 64-state grid. The reference surface is the DCE-estimated conditional-logit model evaluated on the grid attributes (wait $\in \{0, 2, 4, 6\}$, efficacy $\in \{0.3, 0.5, 0.7, 0.9\}$, side-effect level $\in \{0, 1, 2, 3\}$; cash and origin held at reference values). This is a **structural diagnostic** on a model-implied projection surface; the grid extends beyond the directly administered DCE profiles. Spearman’s ρ measures alignment with the DCE-estimated reference; the exact-task categorical validation (Figure 1) is the primary held-out benchmark.

Model	Method	MSE $\downarrow$	MAE $\downarrow$	MVR-Wait (%) $\downarrow$	Spearman’s ρ $\uparrow$
Qwen2.5-72B-Instruct	Raw LLM	0.0258	0.1269	62.5	+0.061
	EFR	0.0016	0.0317	30.2	0.689
DeepSeek V4 Pro	Raw LLM	0.0379	0.1570	34.4	0.302
	EFR	0.0024	0.0392	7.3	0.812
MiroThinker-1.7-mini	Raw LLM	0.0872	0.2644	10.4	0.531
	EFR	0.0054	0.0661	8.3	0.756

4 Experimental Evaluation

4.1 Experimental Setup

We evaluated EFR using three open-source LLMs (Qwen2.5-72B-Instruct, DeepSeek V4 Pro, MiroThinker-1.7-mini; endpoint and access details in Table S2). Decoding parameters were matched across models where the API permitted, with Qwen2.5-72B-Instruct and DeepSeek V4 Pro sampled at temperature = 0.7, top_p = 1.0, max_new_tokens = 512, and $R = 10$ repeated generations per state to average out sampling noise; MiroThinker-1.7-mini was sampled at temperature = 0.0 (deterministic), top_p = 1.0, max_new_tokens = 256, and $R = 10$ repetitions. The empirical choice frontier was estimated from a corrected 6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant on the full Wuhan DCE sample ($N = 1,027$ respondents) on vaccination timing (ISPOR good-research-practices^{13,8}). All LLM evaluations comply with the ViH/ISPOR ELEVATE-GenAI reporting framework (Appendix C).

4.2 Behavioral Consistency with the Empirical Frontier

Table 1 summarizes behavioral consistency on the 64-state evaluation grid. The grid attributes extend beyond the directly administered DCE profiles; the reference surface is a DCE-estimated conditional-logit projection. MVR is the wait-axis monotonicity violation rate; ρ is Spearman’s rank correlation between π and P_{emp} . The exact-task categorical validation (Figure 1) is the primary held-out benchmark. EFR ($\lambda = 0.25$) reduced MSE by 94%, MAE by 75%, and MVR-Wait substantially; all EFR predictions achieved $\rho \approx 0.69$ or higher, exceeding the uniform-noise null. The permutation null (Supplement B) distinguishes models whose LLM component shows detectable DCE-aligned rank information beyond mechanical anchoring (DeepSeek V4 Pro, MiroThinker; $p \leq .003$) from those that do not (Qwen2.5-72B; $p = .179$; evaluated on the local replication that reproduced the canonical $\rho = 0.689$). Alternative-level analysis: Table S7.

Table 2: Behavioral consistency under out-of-design policy scenarios.

Model	Method	MVR ($\downarrow$)	MAD ($\downarrow$)	Spearman ρ ($\uparrow$)
<i>Panel A: Core scenarios (6 paired policy descriptions)</i>				
Qwen2.5-72B-Instruct	Unconstrained LLM	0.333 (2/6)	0.290	0.12
Qwen2.5-72B-Instruct	EFR	0.000	0.072	0.87
DeepSeek V4 Pro	Unconstrained LLM	0.667 (4/6)	0.257	-0.07
DeepSeek V4 Pro	EFR	0.000	0.064	0.87
<i>Panel B: Extended scenarios (14 paired; MVR-Wait defined only for 2 wait-varying pairs)</i>				
Qwen2.5-72B-Instruct	Unconstrained LLM	0.000 (0/2)	0.125	-0.01
Qwen2.5-72B-Instruct	EFR	0.000	0.031	0.93
DeepSeek V4 Pro	Unconstrained LLM	0.000 (0/2)	0.209	-0.06
DeepSeek V4 Pro	EFR	0.000	0.052	0.88

Note. Twenty paired scenarios (40 narrative descriptions total); eight vary waiting time and twelve vary only narrative content. MVR-Wait is reported on the eight wait-varying pairs (Panel A: 6; Panel B: 2); the twelve narrative-only pairs are reported as MAD/ ρ only. MAD is mean absolute deviation from P_{emp} . *Panel B Spearman ρ is reported descriptively only:* because P_{emp} contains only six distinct values across the 28 Panel B scenarios, ρ is strongly affected by ties and lies within the construction-only noise benchmark; it is therefore reported descriptively only.

4.3 Out-of-Design Policy Scenario Evaluation

We evaluated EFR on 20 narrative policy scenarios extending beyond the DCE design (Table 2). Panel A (six pairs) varies waiting time; Panel B (14 pairs) varies other attributes. Full scenario descriptions are provided in the Supplement.

Table 2 summarizes the out-of-design evaluation. In the core scenarios (Panel A), the unconstrained LLMs violated the expected waiting-time ordering in 2 of 6 paired scenarios for Qwen2.5-72B-Instruct and 4 of 6 paired scenarios for DeepSeek V4 Pro. After empirical-frontier regularization, no monotonicity violations were observed for either model, and rank agreement with the empirical behavioral reference increased substantially.

The differences became more pronounced in the extended scenarios (Panel B). Rich textual narratives containing fictional incentives and complex policy descriptions reduced the rank agreement of the unconstrained LLMs to approximately zero for both models. Among the two wait-varying pairs in Panel B, EFR maintained zero monotonicity violations (0/2); the twelve narrative-only pairs are reported as MAD and ρ only (all 20 paired scenarios = 40 narrative descriptions; only Qwen2.5-72B-Instruct and DeepSeek V4 Pro were evaluated on the OOD panel, MiroThinker-

1.7-mini was not). Rank agreement with the empirical reference was $\rho = 0.93$ for Qwen2.5-72B-Instruct and $\rho = 0.88$ for DeepSeek V4 Pro. These out-of-design results characterize structural consistency of the anchored outputs with the model-implied DCE reference surface; they do not establish predictive validity for human behavior outside the observed DCE design.

This experiment evaluates final anchored predictions; the separate contributions of language understanding and empirical regularization are not isolated here.

4.4 Policy Ranking Consistency

EFR increased agreement with the Pure-DCE policy ranking (a structural diagnostic, not an independent validation): Qwen2.5-72B-Instruct Spearman PRC $0.333 \rightarrow 0.867$; DeepSeek V4 Pro $0.133 \rightarrow 0.600$; MiroThinker-1.7-mini $0.533 \rightarrow 0.800$. Because the EFR blend contains 75% P_{emp} by construction, increased agreement with the Pure-DCE ranking is a mechanical consequence of the convex anchor, not an independent behavioral validation. PRC is therefore reported in the supplement as a structural diagnostic only; no significance test is performed on the 15 pairwise comparisons because they share six underlying policy effects and are not 15 independent Bernoulli trials.

5 Discussion

5.1 Principal Findings

EFR improves behavioral consistency without retraining. Three findings emerge. *First*, unconstrained LLMs produce choice probabilities poorly aligned with the DCE-estimated reference: waiting-time monotonicity violations ranged 10.4–62.5% (Qwen2.5-72B: 62.5%; DeepSeek V4 Pro: 34.4%; MiroThinker: 10.4%); rank correlation was weak (ρ : 0.061/0.302/0.531). *Second*, EFR reduced violations to 7.3–30.2% and raised ρ to ≥ 0.69 (0.689/0.812/0.756). Primary exact-task validation ($N = 205, 1,230$ choices) shows multinomial log loss: Pure-DCE 0.911, EFR

0.923, Raw LLM 1.020. EFR improved log loss versus Raw LLM (-0.0967 ; 95% CI -0.1468 to -0.0469); the EFR–Pure-DCE difference was small and imprecise ($+0.0121$; 95% CI -0.0234 to $+0.0472$). Alternative-level analysis ($N = 813$): Table S7. *Third*, policy-ranking agreement increased (PRC: $0.333 \rightarrow 0.867$, $0.133 \rightarrow 0.600$, $0.533 \rightarrow 0.800$); this is a structural diagnostic of the 75% P_{emp} blend, not independent validation (reported in supplement without significance testing).

5.2 Methodological Contributions

Our primary contribution is *inference-time empirical regularization*, a procedure that aligns LLM decisions with empirically estimated behavioral structure without modifying model weights. The design reflects theory-informed machine learning: established empirical models provide structural priors for generative AI systems. EFR is a wrapper, not a model: it can be applied to any LLM (closed-source or open-source) and any empirical choice frontier (mixed logit, latent class, or non-parametric). The convex combination $\pi_{\text{EFR}} = \lambda \pi_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$ is the simplest possible anchor; future work could explore non-convex anchors (e.g., mixture-of-experts gating, Bayesian shrinkage toward P_{emp} priors).

A robustness stress test on eight Panel B scenarios across three smaller open-source models (4,800 prompts) corroborated this view: Qwen2.5-7B produced constant outputs (zero variance, default-answer signature), DeepSeek V4 (Flash variant) achieved 0% parse on the waiver-language N1 scenario. *These failures are a safety property, not evidence of LLM value-add*: when LLM output is degenerate, EFR continues to assign 75% of the weight to P_{emp} , bounding—but not eliminating—the LLM component (EFR is not a reliability gate, and the $\lambda = 0.25$ weight is fixed rather than adaptive). Our use of LLMs as anchored narrative components differs from prior tool-oriented LLM work: the empirical frontier, not the LLM, supplies the behavioral signal; the LLM component is permitted only in the convex blend with P_{emp} .

5.3 Implications for Health Policy Analysis

The EFR framework has three HTA implications. *First*, a recent DCE unlocks a low-marginal-cost narrative-policy simulation pipeline: HTA agencies need only the LLM API, DCE estimates, and a deterministic blend. This may be useful for exploratory screening when immediate collection of a new DCE is impractical. *Second*, EFR extends the productive life of an existing DCE: EFR applies to narrative policy descriptions (fictional incentives, side-effect profiles, communication strategies) outside the administered structured DCE profiles. *Third*, the framework is best suited to early-stage screening, not high-stakes final decisions; for the latter, EFR outputs should be triangulated with traditional stated-preference methods.

Limitations and design considerations. (i) *Wuhan DCE, single-jurisdiction sample.* The reference surface is estimated from one DCE fielded in a single Chinese city (collection period and full DCE questionnaire reported in the public code repository at <https://github.com/REDACTED/behavioral-digital-twins>); preference stability across pandemic phases is not established. Replication across additional jurisdictions and time periods is planned future work (Conclusion). (ii) *Conditional-logit specification and frontier stability.* The static frontier uses a 6-parameter grouped/conditional logit (five attribute coefficients plus an opt-out alternative-specific constant) on the full DCE sample, not a random-coefficients mixed logit. Respondent-cluster bootstrap (2,000 replicates) confirms frontier stability: Spearman ρ between bootstrap and canonical frontier has mean 0.988 (median 0.993; 95% CI [0.948, 1.000]) and all six coefficients retain their sign in $\geq 97.7\%$ of replicates (Appendix J of the Supplementary Material). The corrected opt-out ASC narrows P_{emp} to [0.47, 0.64]; we trade within-DCE-design contrast for a behaviorally honest specification. (iii) $\lambda = 0.25$ as an illustrative conservative operating point. The blending weight is a fixed illustrative conservative operating point preserving a non-zero LLM contribution while keeping the empirical DCE dominant; it is not a test-set-optimized predictive optimum. The held-out DCE log loss and Brier score are reported as sensitivity diagnostics and were not used to select λ . Structurally, this weight retains only one quarter of the raw LLM–frontier deviation pointwise, me-

chanically bounding frontier-relative squared deviation to $\lambda^2 = 6.25\%$ of the unconstrained LLM value (Supplementary Material F.2). **The paper’s substantive conclusions derive from the full λ sensitivity analysis (Supplementary Material F.3), not from the specific value $\lambda = 0.25$.** (iv) *Inference-time regularization vs. retraining.* EFR is a convex combination of two streams, not a learned gate; non-convex anchors (mixture-of-experts gating, Bayesian shrinkage toward P_{emp} priors) remain future work.

5.4 Why Not Use the Discrete Choice Model Alone?

EFR trades exact-task predictive performance for narrative-policy evaluation capability (multinomial log loss: Pure-DCE 0.911; EFR 0.923; Raw LLM 1.020; uniform benchmark 1.099). **EFR materially improved exact-task predictive performance relative to the unconstrained LLM** (9.5% reduction in log loss) and approached the predictive performance of the Pure-DCE model (within 1.3%), while retaining a non-zero LLM component for narrative extension. EFR does not outperform the DCE model within the original structured choice domain, but it substantially improves on the unconstrained LLM and recovers most of the DCE model’s predictive performance while preserving an LLM component for narrative extension. The NDS pilot ($n = 2$) is reported as illustrative mechanism evidence only: isotonic NDS $\rho = -0.943$ vs. EFR $\rho = +0.894$ for DeepSeek V4 Pro. **Figure 1** reports the exact-task categorical held-out validation results.

5.5 Implementation and Policy Recommendations

Significance. EFR is a constraint wrapper that applies an existing DCE-derived frontier to narrative-policy evaluation. The fixed $\lambda = 0.25$ anchor bounds, rather than eliminates, the deviation of LLM outputs from the DCE-derived frontier. The OOD narrative evaluation demonstrates that EFR preserves structural consistency with the DCE-estimated reference (monotonicity, rank order), but this is a constraint effect, not evidence that the LLM component provides accurate behavioral simulation for novel policy scenarios. EFR is best understood as a narrative-extension wrapper for an existing DCE, not as a replacement for the underlying DCE model or as validation of LLMs as



Figure 1: Independent local replication validation multinomial log loss on the six administered DCE choice tasks (Pure-DCE: 0.911; EFR: 0.923; Raw LLM: 1.020; uniform benchmark: 1.099; $N = 205$ held-out respondents, 1,230 choices; local Ollama Qwen2.5-72B). In respondent-level paired bootstrap analyses, EFR improved log loss relative to Raw LLM (difference -0.0967 ; 95% CI, -0.1468 to -0.0469), whereas the EFR–Pure-DCE difference was small and imprecisely estimated ($+0.0121$; 95% CI, -0.0234 to $+0.0472$). Because all respondents completed identical tasks, this validation evaluates generalization to unseen respondents rather than to unseen choice tasks or attribute profiles.

standalone behavioral simulators. Implementation requires no model retraining: an analyst estimates a 6-parameter grouped/conditional logit from an existing DCE, generates LLM predictions, and blends them through $\pi_{\text{adj}} = \lambda \pi_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$ ($\lambda = 0.25$). Three policy implications follow. *With a DCE*, EFR provides a path from evidence to narrative policy simulation. *Without a DCE*, the framework is not a substitute for stated-preference data. *Best for early-stage screening*, not high-stakes final decisions. The permutation analysis indicates that incremental DCE-aligned rank structure in the LLM component is model-dependent: it was detectable for DeepSeek V4 Pro and MiroThinker, but not for Qwen2.5-72B.

5.6 Conclusion

EFR anchors LLM outputs to DCE estimates without retraining. Independent local replication validation on the six administered DCE choice tasks ($N = 205$ respondents, 1,230 choices; local Ollama Qwen2.5-72B) shows multinomial log loss: Pure-DCE (0.911), EFR (0.923), Raw LLM (1.020), Uniform (1.099). EFR materially improved exact-task predictive performance relative to the unconstrained LLM (9.5% reduction in log loss) and approached the predictive performance of the Pure-DCE model (within 1.3%), while retaining a non-zero LLM component for narrative extension. EFR does not outperform the DCE model within the original structured choice domain, but it substantially improves on the unconstrained LLM and recovers most of the DCE model’s predictive performance while preserving an LLM component for narrative extension. On the structured 64-state grid, MVR-Wait fell from 10.4–62.5% to 7.3–30.2% across three LLMs (Qwen2.5-72B-Instruct: 62.5% $\rightarrow$ 30.2%; DeepSeek V4 Pro: 34.4% $\rightarrow$ 7.3%; MiroThinker-1.7-mini: 10.4% $\rightarrow$ 8.3%) under EFR anchoring at $\lambda = 0.25$ (Table 1). *In the eight wait-varying OOD narrative pairs (40 narrative descriptions total, evaluated with Qwen2.5-72B-Instruct and DeepSeek V4 Pro only), EFR eliminated the observed waiting-time reversals on Panel A (Qwen: 2/6 to 0/6; DeepSeek: 4/6 to 0/6) and held the Panel B wait-varying pairs at 0/2 for both models; the twelve narrative-only pairs (varying side-effect, efficacy, or fictional narrative content while holding wait constant) are reported as MAD and Spearman ρ only.* EFR is best understood as a constrained narrative-extension mechanism for an existing DCE, with the LLM component permitted only inside the convex blend with P_{emp} . The empirical reference, not the LLM, supplies the behavioral signal; whether the narrative extension is behaviorally more accurate on out-of-design scenarios remains to be established by future work with human ground truth. Three directions are planned future work: (i) replication across additional diseases, jurisdictions, and policy domains; (ii) exploration of non-convex anchoring mechanisms (e.g., mixture-of-experts gating); (iii) quasi-experimental validation with human ground truth on out-of-design narrative scenarios; and (iv) dynamic DCE updating, in which the empirical frontier is re-estimated and the blending weight re-calibrated as new stated-preference data accumulate.

References

- [1] Gati Aher, Rosa I. Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies, 2022.
- [2] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023.
- [3] Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S. Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- [4] John F. P. Bridges, A. Brett Hauber, Deborah Marshall, Andrew Lloyd, Lisa A. Prosser, Dean A. Regier, F. Reed Johnson, and Josephine Mauskopf. Conjoint analysis applications in health—How are studies being designed and reported?: An update: Current practices in ISPOR conjoint analysis applications. *Value in Health*, 16(2):287–298, 2013. doi: 10.1016/j.jval.2012.12.012.
- [5] Amy Hai Yan Chan et al. Vaccine decision making in new zealand: A discrete choice experiment. *BMC Public Health*, 24:447, 2024. doi: 10.1186/s12889-024-17965-5.
- [6] Tina L. Cheng, Juan Marcos Gonzalez, Matthew M. Engelhard, Shelby D. Reed, and Semra Ozdemir. Can large language models predict patient treatment choices? A discrete choice experiment framework. *Value in Health*, 29(5), 2026. doi: 10.1016/j.jval.2026.04.006. Published online May 5, 2026.
- [7] Lucinda Gullison and Feng Fu. Working with large language models to enhance messaging effectiveness for vaccine confidence. *arXiv preprint arXiv:2504.09857*, 2025.

- [8] A. Brett Hauber, Juan Marcos González, Catharina G. M. Groothuis-Oudshoorn, Thomas Prior, Deborah A. Marshall, Charles Cunningham, Maarten J. IJzerman, and John F. P. Bridges. Statistical methods for the analysis of discrete choice experiments: A report of the ISPOR conjoint analysis good research practices task force. *Value in Health*, 19(4):300–315, 2016. doi: 10.1016/j.jval.2016.04.004.
- [9] David A. Hensher and William H. Greene. The mixed logit model: The state of practice. *Transportation*, 30(2):133–176, 2003.
- [10] David A. Hensher, John M. Rose, and William H. Greene. *Applied Choice Analysis: A Primer*. Cambridge University Press, Cambridge, 2nd edition, 2015.
- [11] Arend Hintze, Charu Bisht, Jory Schossau, and Ralph Hertwig. Nonlinear transformation of probabilities by large language models. *Computers in Human Behavior: Artificial Humans*, 6:100227, 2025. doi: 10.1016/j.chbah.2025.100227.
- [12] John J. Horton. Large language models as simulated economic agents: What can we learn from homo silicus? *NBER Working Paper*, (w31122), 2023. doi: 10.3386/w31122.
- [13] F. Reed Johnson, Esther Lancsar, Deborah Marshall, Vikram Kilambi, Andreas Muehlbacher, Dean A. Regier, Brian W. Bresnahan, Brett J. Kanninen, and John F. P. Bridges. Constructing experimental designs for discrete-choice experiments: Report of the ISPOR conjoint analysis good research practices task force. *Value in Health*, 16(1):3–13, 2013. doi: 10.1016/j.jval.2012.08.223.
- [14] Fatima Koaik, Aayush Gupta, and Farahan Raza Sheikh. Llm powered social digital twins: A framework for simulating population behavioral response to policy interventions. arXiv preprint arXiv:2601.06111 [cs.AI], January 2026. 13 pages, 1 figure, 4 tables; available at <https://arxiv.org/abs/2601.06111>.
- [15] Yiming Li and Dacheng Tao. Position: AI agents are not (yet) a panacea for social simulation, February 2026.

- [16] Jordan J. Louviere, David A. Hensher, and Joffre D. Swait. *Stated Choice Methods: Analyzing and Interpreting Data*. Cambridge University Press, Cambridge, 2000.
- [17] Nicholas Luo, Stephanie Tan, Esha Wadhwa, and Jiraganya Pwu. Simulating sponsor-HTA conversations: A role-based multi-agent large language model framework. *Value in Health*, 28(12):S1–S6, 2025. Conference abstract P4; multi-agent LLM framework for sponsor-HTA interactions.
- [18] Xinyi Mu, Xuanwen Ding, Qi He, Liang Wang, Jingcong Liang, Xinnong Zhang, Libo Sun, Jiayu Lin, Jie Zhou, Xuanjing Huang, and Zhongyu Wei. From individual to society: A survey on social simulation driven by large language model-based agents. *arXiv preprint arXiv:2412.03563*, 2024.
- [19] Joon Sung Park, Joseph O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023. doi: 10.1145/3586183.3606763.
- [20] Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? In *International Conference on Machine Learning (ICML)*, 2023.
- [21] Neil K. R. Sehgal, Sunny Rai, Manuel Tonneau, Anish K. Agarwal, Joseph Cappella, Melanie Kornides, Lyle Ungar, Alison Buttenheim, and Sharath Chandra Guntuku. Conversations with ai chatbots increase short-term vaccine intentions but do not outperform standard public health messaging. *arXiv preprint arXiv:2504.20519*, 2025.
- [22] Akshay Shree, Oliver Phillips, Ryan Sullivan, Pablo Garcia, and Aarti Bansal. Stability of a large language model for data extraction in systematic reviews: A sensitivity analysis. *Value in Health*, 28(12):1832–1839, 2025. Conference abstract MSR105; full sensitivity analysis by varying time of day and geographic location.

- [23] Vikas Soekhai, Esther W. de Bekker-Grob, Alan R. Ellis, and Caroline M. Vass. Discrete choice experiments in health economics: Past, present and future. *PharmacoEconomics*, 37(2):201–226, 2019. doi: 10.1007/s40273-018-0734-2.
- [24] Petter Törnberg and Marlon Larooij. Validation is the central challenge for generative social simulation: A critical review of LLMs in agent-based modeling. *Artificial Intelligence Review*, 59(1):15, 2025. doi: 10.1007/s10462-025-11412-6.
- [25] Olivier Toubia, Junchao Yang, Deepak Pai, Youngseok Chae, and Jonah Berger. Twin-2K-500: A data set for building digital twins of US adult consumers from their Amazon purchase histories. *Marketing Science*, 2025. Forthcoming; preprint cited via DOI 10.1287/mksc.2025.0262.
- [26] Kenneth E. Train. *Discrete Choice Methods with Simulation*. Cambridge University Press, 2 edition, 2009.
- [27] Emma Von Hoene, Amira Roess, Hamdi Kavak, and Taylor Anderson. Synthetic population generation with public health characteristics for spatial agent-based models. *PLOS Computational Biology*, 2025. doi: 10.1371/journal.pcbi.1012439.
- [28] Zhengyu Wang, Tomasz Pawlak, and Konstantinos Sivakumar. Heating-choice stated preferences with large language models. *Working Paper*, 2025.
- [29] Zhengyu Wang et al. Can LLMs simulate human responses? *arXiv preprint arXiv:2503.10652*, 2025.

Supplementary Materials

The following supplementary materials accompany this manuscript and are available in the online appendix.

A. Tables

- **Table S1.** Observed EFR Spearman ρ versus uniform-noise and permutation nulls on the 64-state grid.
- **Table S2.** LLM decoding and endpoint parameters used in the main, supplementary, and stress-test evaluations.
- **Table S3.** *Administered* DCE attribute levels: waiting time $\{0, 1, 3, 6\}$ months (4 real design levels; the raw coded data contain a single 1-row outlier at wait = 2 months from Respondent 1 only, which was excluded from all analyses before sample-size calculations). The analytic sample contains 18,485 alt-rows (6,162 choice tasks $\times$ 3 alternatives minus 1 anomalous row). All 1,027 respondents completed the same six choice tasks; task presentation order was randomized across respondents to mitigate order effects. Each task presented two non-opt-out alternatives and an opt-out alternative. The held-out validation subsample ($N = 205$ respondents) contains 1,230 valid tasks (205×6 ; the anomalous row was from Respondent 1, who was not in the held-out subsample). The design therefore contained six unique administered choice tasks and 12 non-opt-out profiles per respondent (12 unique non-opt-out profiles in the design; $12,324 = 1,027 \times 12$ profile instances across all respondents). The 12 administered non-opt-out attribute combinations per respondent enumerate 12 of the $3 \times 3 \times 3 \times 2 \times 4 = 216$ non-opt-out candidate cells (5.6% per-respondent coverage).
- **Table S4.** Attribute coding dictionary: DCE questionnaire display, estimation coding, 64-state evaluation-grid coding, and coefficient unit definitions.
- **Table S5.** Conditional-logit estimates (corrected 6-parameter specification) used to construct the empirical choice frontier. Standard errors from BFGS Hessian; bootstrap 95% confidence intervals for the 6 coefficients.

- **Table S6.** λ -sensitivity for frontier-alignment diagnostics on the 64-state grid. Pure-DCE baseline, unconstrained LLM, and EFR Anchor evaluated across $\lambda \in \{0.00, 0.10, 0.20, 0.25, 0.30, 0.40, 0.50\}$.
- **Table S7.** λ -sensitivity for held-out DCE predictive validation on the 813 *alternative-level* rows subset (from 205 held-out respondents). Alt-level binary log loss, Brier score, and AUC across λ .
- **Table S8.** Data-scaling ablation: EFR Anchor ($\lambda = 0.25$) vs. unconstrained LLM across training fractions (30%, 50%, 70%, 100%).
- **Table S9.** Parse success rates across models and experiment scales.
- **Table S10.** Runtime summary for full-scale unconstrained LLM generation.
- **Table S11.** Policy Ranking Consistency (PRC) relative to the Pure-DCE reference; descriptive structural diagnostic, no significance test.
- **Table S12.** Respondent-level paired cluster bootstrap for exact-task multinomial log loss (six canonical tasks; $N = 205$ respondents).
- **Table S13.** Bootstrap intervals for selected frontier-alignment diagnostics.
- **Table S14.** Attribute-level mapping between the administered DCE design and the 64-state LLM evaluation grid. The matched subset is restricted to cells where the DCE design level and the grid level coincide as a float value (no rescaling, no interpolation).
- **Table S15.** Covariate balance between matched and unmatched held-out subsets.
- **Table S16.** Primary exact-task multinomial held-out log loss on six canonical DCE tasks ($N = 205$ respondents, 1,230 choices). Pure-DCE, EFR, Raw LLM, and uniform baseline.

B. Figures

- **Figure S1.** Data-scaling ablation: EFR Anchor ($\lambda = 0.25$) evaluated across DCE training shares (30%, 50%, 70%, 100%); companion to the data-scaling narrative in Appendix G.

C. Replication Materials

Replication materials are available at <https://github.com/REDACTED/behavioral-digital-twins>, including: the questionnaire, code, de-identified aggregate/grid-level inputs, model outputs, and derived analysis datasets required to reproduce the reported LLM and frontier-alignment analyses. Individual-level DCE responses are not publicly released because of participant confidentiality.